Release note

Topic **HDG 2.01 firmware for ZED-X20D-00B**

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1 General information

1.1 Scope

This Release note applies to the u-blox HDG 2.01 firmware version for the ZED-X20D-00B heading module only.

1.2 Open-source declaration

The u-blox positioning product described in this Release note, comprising the company's proprietary software, does not contain open-source software to declare.

1.3 Related documentation

[1] ZED-X20D Data sheet, UBXDOC-304424225-20103
[2] ZED-X20D Integration manual, UBXDOC-304424225-21476
[3] u-blox X20 HDG 2.01 Interface description, UBXDOC-304424225-21635
[4] u-blox EVK-X20D User guide, UBXDOC-304424225-21353
[5] u-center 2 GNSS evaluation software for Windows, version 26.10.1 or later version, https://www.u-blox.com/en/u-center-2
[6] u-blox GPS L5 configuration Application note, UBX-21038688

2 Released firmware image

Released firmware image	Description
File	UBX_20_HDG201.fbbf646a0046481eb41cf3d1386f83b7.bin
Firmware version	EXT HDG 2.01 (f4beec9) FWVER=HDG 2.01
Protocol version	57.03
ROM base support	ROM BASE 0x00A9D329

2.1 Released software tools

2.1.1 u-center 2

Use u-center 2 version 26.10.1 or later with the ZED-X20D HDG 2.01 firmware.

3 Firmware description

The key features supported by HDG 2.01 (only for ZED-X20D) are:

- High precision positioning output (PVT, as in ZED-X20P)
- Dual all-band antenna support
- Dual output of raw code and carrier phase measurement for all supported GNSS signals
- Dual-antenna, all-band high-precision GNSS-based heading solution
- RTK, PPP-RTK and PPP functionality
    - PointPerfect correction service
    - Galileo High Accuracy Service (HAS)

3.1 Supported GNSS signals

The HDG 2.01 firmware supports the following GNSS signals:

- GPS: L1C/A, L2C, L5
- Galileo: E1B/C, E5a, E6
- BeiDou: B1I, B1C, B2a, B3I
- GLONASS: L1OF, L2OF
- QZSS: L1C/A, L1C/B, L1S, L2C, L5
- NavIC: L5
- SBAS: L1C/A

☞ Note that the GPS L5 signals are flagged as unhealthy at the time of this release. The receiver does not use unhealthy signals for navigation by default. See the GPS L5 configuration application note [6] for further details.

☞ Regional constellations, such as QZSS and NavIC, are disabled by default. They can be enabled on demand.

☞ GLONASS Geo-blocking feature: GLONASS is generally available but restricted in some geographical areas.

3.2 GNSS and heading features

- Dual output of code and carrier phase measurements (UBX-RXM-RAWX)
- High precision heading (dual antenna)
- AssistNow Live Orbits
- Geofencing with support for location restriction and anti-theft functionality through navigation messages
- Monitoring and management functions for embedded system health, including I/O queues, pin status, and correction status
- Personalization with individualized identity for access to high value features and subscriptions
- Hardware backup modes
- Anti-jamming capabilities, including RF interference and jamming detection and reporting
- Anti-spoofing capabilities, including spoofing detection and reporting
- Galileo Open Service Navigation Message Authentication (OSNMA)
- Configuration lockdown, allowing receiver configuration to be locked by command
- Secure boot, enabling execution of only signed firmware images
- Secure storage with tamper-resistant, end-to-end security
- RTK high-precision positioning
- PPP-RTK high-precision positioning
- PPP positioning solution using the Galileo High Accuracy Service

3.3 Message interface

For a detailed description of the message interfaces available for the ZED-X20D HDG 2.01 release, see the Interface description [3].

3.3.1 NMEA

The default NMEA version is 4.11. Alternatively, versions 4.10 and 4.0 can also be enabled.

3.3.2 UBX

This firmware supports the UBX protocol version 57.03. For further information, refer to the Interface description [3].

3.3.3 RTCM

The released firmware supports the RTCM3 standard up to version 3.4.

3.3.4 SPARTN

The released firmware supports the SPARTN protocol up to version 2.0.2.

4 Changes and improvements

Compared to the HDG 2.00 firmware, the changes/improvements in HDG 2.01 are:

- **Improved HAS performance and robustness,** including correction continuity and handling of invalid signal.
- **Improved QZSS L1C/B signal tracking** and corrected a pseudorange measurement bias.
- **Improved spoofing-mitigation behavior at navigation update rates above 1 Hz.** Under certain jamming and spoofing-like conditions, the receiver could switch between a valid fix and NO FIX.
- **Improved RTK performance** when ionosphere estimation (CFG-NAVHPG-IONO_EST) was disabled.
- **Added GGA message output on UART2 interface**

4.1 Interface changes

For a detailed description of the interfaces available in this firmware, see the u-blox X20 HDG 2.01 Interface description [3]. This section lists only the interface changes compared to HDG 2.00.

4.1.1 New

Message	Audience	Description / Comment
CFG-MSGOUT-NMEA_ID_GGA_UART2	PUB	Added. Enables NMEA GGA output message on the UART2 To enable GGA output on UART2, set config item CFG-MSGOUT-NMEA_ID_GGA_UART2 to true.

5 Known limitations

5.1 Interface

UART2 is mainly an input-only interface that, with the HDG 2.01 firmware, supports the reception of RTCM and SPARTN correction data.

GGA is the only supported output message on the UART2 with this firmware.

5.2 NMEA in u-center 2

Disabling or enabling NMEA child messages in u-center 2 affects only GGA and GLL messages.

Workaround: each NMEA child message has to be individually disabled or enabled.

5.3 Software standby mode

The SW backup feature is not functional with the HDG 2.01 firmware.

5.4 Update rate

ZED-X20D heading module supports dual antenna. For optimal performance, the update rates on antenna 1 and antenna 2 should be identical. If the application requires the update rate to differ between antenna 1 and antenna 2, it is possible with technical support from u-blox*, but to ensure reliable performance, it is recommended to have antenna 1 update rate as an integer multiple of the antenna 2 update rate, as in the examples below:

- 20 Hz Navigation update rate at antenna 1, 10 Hz / 5 Hz heading at antenna 2
- 25 Hz Navigation update rate at antenna 1, 12.5 Hz / 5 Hz heading at antenna 2

** With the HDG 2.01 firmware, contact u-blox local technical support for further information and guidelines about special configuration of update rate.*

5.5 UBX-MON-* messages

Only the UBX-MON messages below have valid content for HDG 2.01 firmware:

- UBX-MON-VER
- UBX-MON-RF
- UBX-MON-SPAN
- UBX-MON-GNSS
- UBX-MON-PATCH

5.6 NMEA-TXT

Some NMEA-TXT information messages can be missing. UBX-INF messages can be used as a workaround.

5.7 Galileo

Galileo HAS and Galileo OSNMA cannot be enabled at the same time.